

# Morning Paper Report - Friday, September 18, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

## 1. Uncertainty-Aware Egocentric Activity Anticipation from Long-Horizon Daily Video Context

Frida Eklund

Global Media and Social Sciences Research Journal, 2026

Source: Semantic Scholar; matched terms: modeling, deep learning, egocentric

Link: <https://www.semanticscholar.org/paper/9039ca6d655c23a74bf4b62d07c0fbbafe49e6d9>

**Goal:** Anticipating future human activities from first-person visual data represents a foundational challenge in computer vision and human-computer interaction

**Method:** By modeling the inherent ambiguity in future predictions, the proposed framework captures the diverse range of plausible future actions while maintaining a robust representation of past and present visual cues

**Results:** Anticipating future human activities from first-person visual data represents a foundational challenge in computer vision and human-computer interaction

**Conclusion:** The proposed framework is evaluated against deterministic baselines, highlighting its superior capability to handle the stochastic nature of human routines. Ultimately, this research provides a scalable and robust approach for integrating predictive vision systems into autonomous agents and wearable cognitive assistants, laying the groundwork for more intelligent, context-aware technological interventions.

## 2. Predictive coding yields a brain-like code for shape from motion

Yoon Bai, Thomas P. O'Connell, Yoni Friedman, Ani Ayzarian-Hancock, Hannah Maver, Joshua B. Tenenbaum, et al.  
bioRxiv, 2026

Source: Semantic Scholar; matched terms: neural network, neural networks, predictive coding, natural video

Link: <https://www.semanticscholar.org/paper/f1da4a74eeebd8162ca2c616a957e09d863df9a5>

**Goal:** Interacting with the world relies on knowing its three-dimensional structure, yet vision receives a stream of two-dimensional images

**Method:** Only predictive coding video models—trained to fill in missing content in natural videos—reproduced both neuronal responses and human behavior

**Results:** Recovering 3D geometry from dynamic input is therefore foundational to vision, but lacks a computational account

**Conclusion:** Recovering 3D geometry from dynamic input is therefore foundational to vision, but lacks a computational account

## 3. Encode-Store-Retrieve: Augmenting Human Memory through Language-Encoded Egocentric Perception

Junxiao Shen, John J. Dudley, P. O. Kristensson

International Symposium on Mixed and Augmented Reality, 2023

Source: Semantic Scholar; matched terms: human memory, egocentric

Link: <https://www.semanticscholar.org/paper/4ee893edd5411d0615c38296232ff4b8e02d7378>

**Goal:** We depend on our own memory to encode, store, and retrieve our experiences

**Method:** This approach harnesses the power of large vision language models to perform the language encoding process

**Results:** Our agent underwent extensive evaluation using the QA-Ego4D dataset and achieved state-of-the-art results with a BLEU score of 8.3, outperforming conventional machine learning models that scored between 3.4 and 5.8

**Conclusion:** The results of this study show that the agent results in significantly better recall performance on episodic memory tasks compared to human participants. The results also highlight the agent's practical applicability and user acceptance.

## 4. Working Memory in Recurrent Spiking Neural Networks With Heterogeneous Synaptic Delays

Laurent U Perrinet

Lecture Notes in Computer Science, 2027

Source: Crossref; matched terms: working memory, neural network, neural networks

Link: [https://doi.org/10.1007/978-3-032-38398-3\\_22](https://doi.org/10.1007/978-3-032-38398-3_22)

**Goal:** The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

**Method:** Method details were not available in the retrieved metadata.

**Results:** Result details were not available in the retrieved metadata.

**Conclusion:** Open the linked record to inspect the full abstract or paper before journal-club use.

## 5. Modelling working memory in neuron-astrocyte network

Yuliya Tsybina, Susan Gordleeva, Mikhail Krivososov, Innokentiy Kastalskiy, Alexey Zaikin, Alexander Gorban

2021 International Joint Conference on Neural Networks (IJCNN), 2021

Source: Crossref; matched terms: working memory, modelling, neural network, neural networks

Link: <https://doi.org/10.1109/ijcnn52387.2021.9533307>

**Goal:** The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

**Method:** Method details were not available in the retrieved metadata.

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